

SB STOCKS

MERN Based Paper Trading Platform

Project Documentation

Submitted by:

Team Members:

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5. _____ - _____

Internship: SmartBridge Virtual Internship 2026

1. Project Overview

SB Stocks is a MERN stack based paper trading platform that allows users to practice stock trading using virtual money. The application provides users with access to US stock market information, historical stock trends, and portfolio management features.

The platform enables users to explore different stocks, analyze price movements, simulate buying and selling of stocks, and track their investment performance without using real money.

2. Problem Statement

Stock trading involves financial risks, and beginners often hesitate to enter real markets without proper knowledge and experience. There is a need for a risk-free environment where users can understand stock market operations and practice trading strategies.

The proposed system provides a paper trading environment where users can learn and practice stock trading using virtual funds.

3. Objectives

The main objectives of this project are:

- To develop a stock paper trading platform using MERN stack.
 - To provide users with stock market information.
 - To display historical stock price trends.
 - To simulate buying and selling of stocks using virtual money.
 - To maintain user portfolio and transaction records.
 - To help users understand stock trading concepts without financial risk.
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4. Proposed System

The proposed system is a web-based paper trading application where users can:

- Create an account and login.
 - View available US stocks.
 - Analyze stock price trends.
 - Buy and sell stocks virtually.
 - Manage their portfolio.
 - Track their trading history.
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5. Features

5.1 User Authentication

Features:

- User registration
- User login
- Secure user access

5.2 Stock Market Dashboard

Features:

- Display stock listings
- Search stocks
- View current stock prices

5.3 Stock Analysis

Features:

- View stock details
- Display historical price charts
- Analyze market trends

5.4 Paper Trading

Features:

- Virtual wallet balance
- Buy stocks
- Sell stocks
- Update portfolio automatically

5.5 Portfolio Management

Features:

- View owned stocks
- Track investment value
- Calculate profit/loss

5.6 Transaction History

Features:

- Store previous trades
- Display buying and selling records

6. Technology Stack

Component	Technology
Frontend	React.js
Backend	Node.js, Express.js
Database	MongoDB

API Integration Stock Market API

Charts Chart Library

Version Control GitHub

7. System Architecture

The system follows a three-tier architecture consisting of frontend, backend, and database layers.

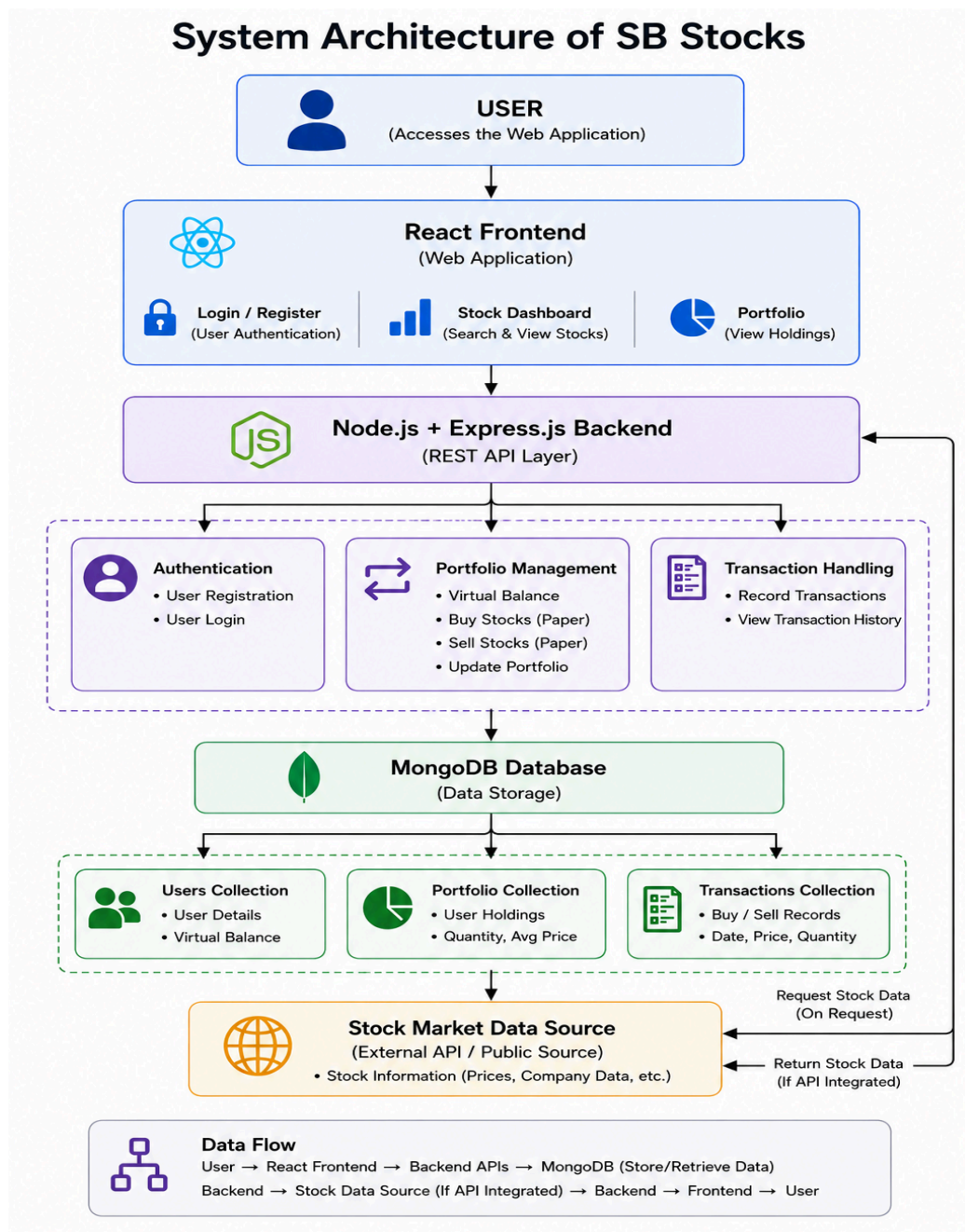
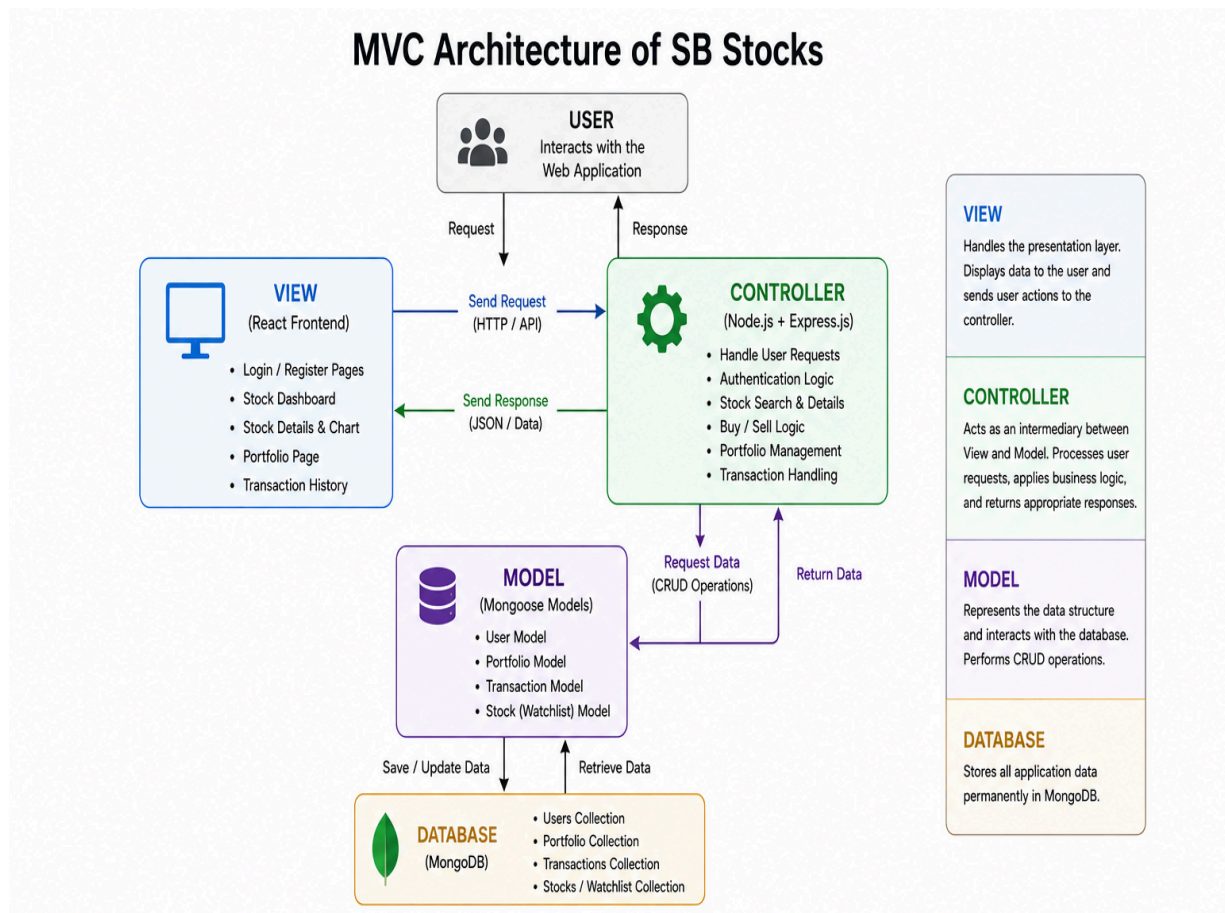


Diagram explanation:

- React.js is used for developing the user interface.
- Node.js and Express.js handle backend operations and API requests.
- MongoDB stores user information, portfolio data, and transaction details.
- Stock Market API provides stock information and historical data.

8.MVC Architecture Design



9. Database Design

The application uses MongoDB for storing application data.

Main collections:

User Collection

Attributes:

- User ID
- Name
- Email
- Password
- Virtual Balance

Stock Collection

Attributes:

- Stock ID
- Stock Symbol
- Company Name
- Stock Price

Portfolio Collection

Attributes:

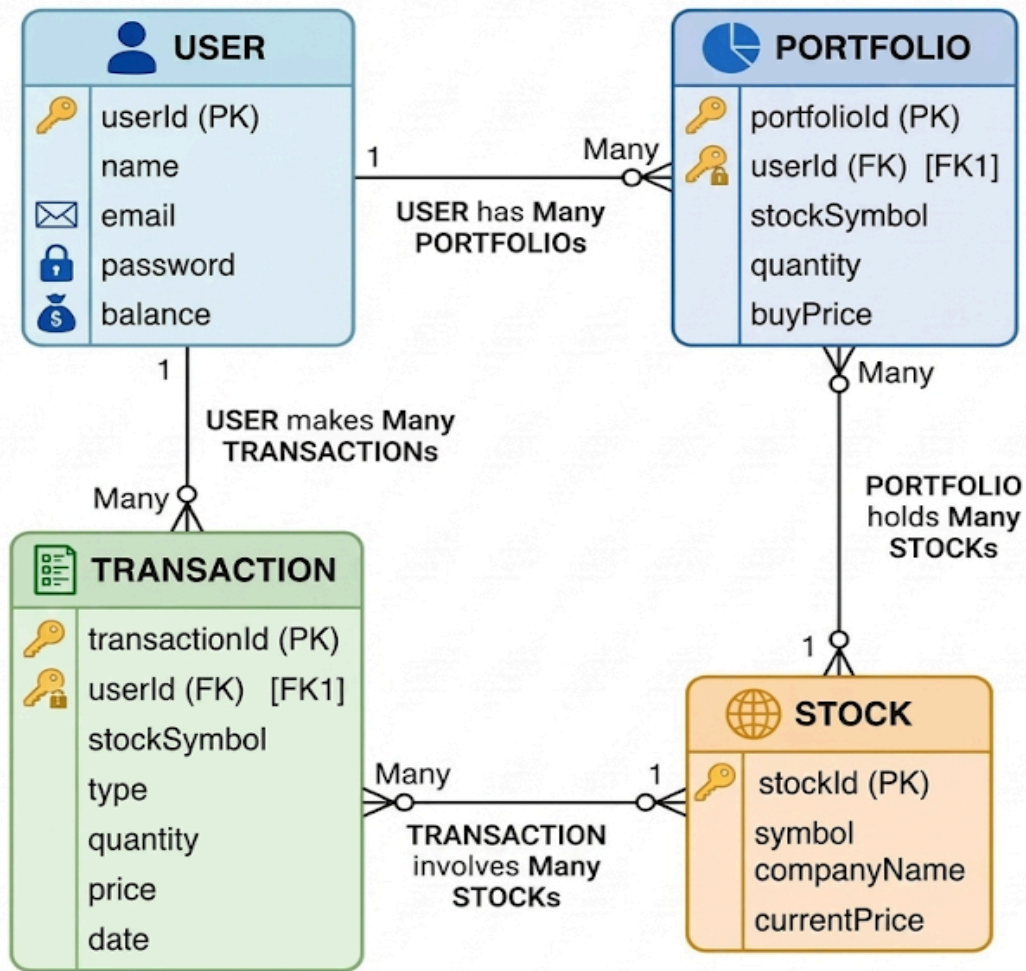
- Portfolio ID
- User ID
- Stock Symbol
- Quantity
- Investment Value

Transaction Collection

Attributes:

- Transaction ID
- User ID
- Stock Symbol
- Transaction Type
- Quantity
- Price
- Date

Database Entity-Relationship (ER) Diagram of SB Stocks



Key



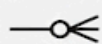
(PK) = Primary Key

[FK1] references USER.userId



(FK) = Foreign Key

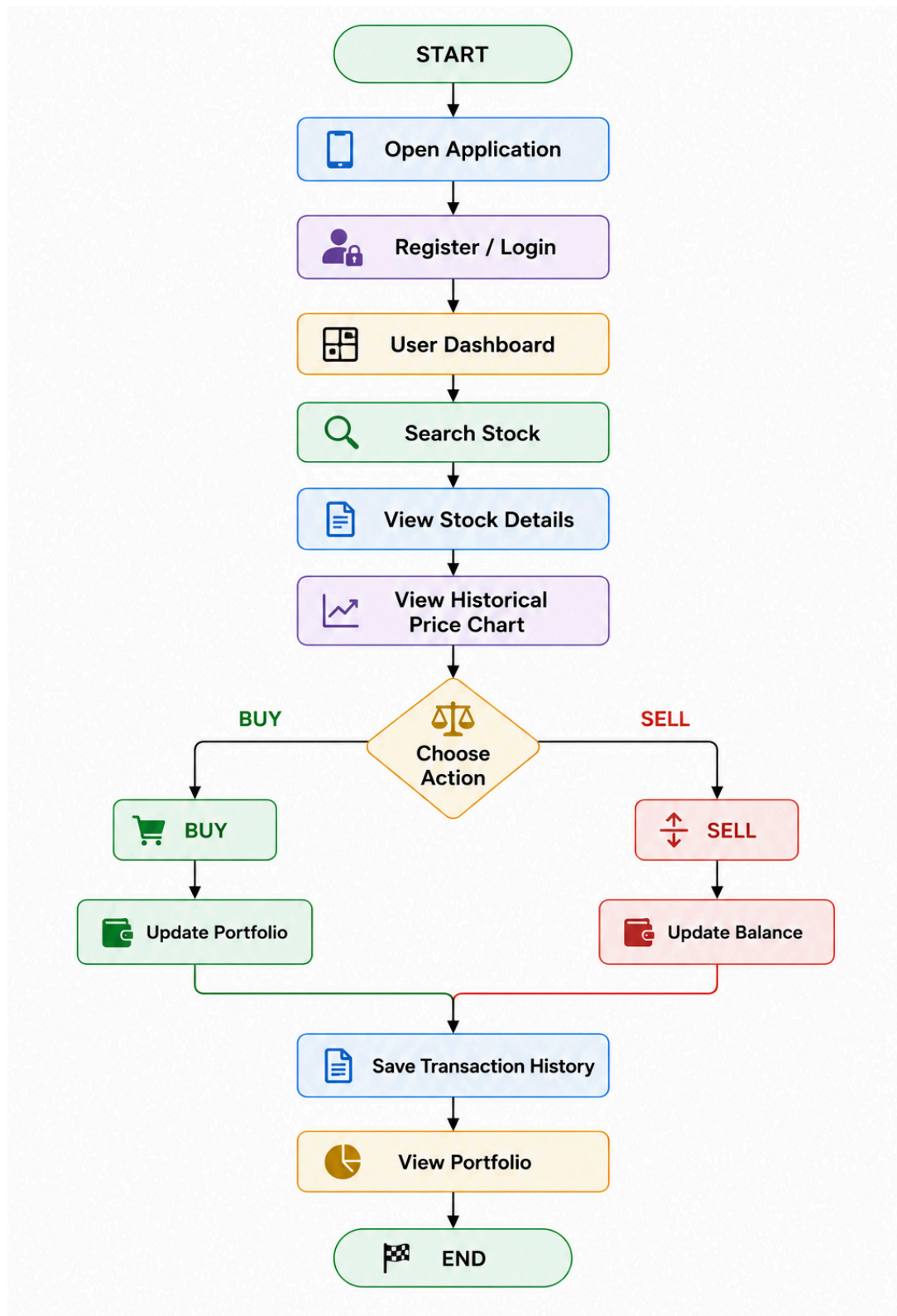
[FK2] references STOCK.symbol



Relationship Type

1-to-Many Crow's Foot

10. User Flow



11. Module Description

Authentication Module

Responsible for managing user registration and login.

Stock Module

Responsible for fetching and displaying stock market information.

Trading Module

Handles virtual buying and selling of stocks.

Portfolio Module

Displays user-owned stocks and investment performance.

Transaction Module

Maintains trading records.

12. Roles and Responsibilities

Member 1: _____

Role: Frontend Developer

Responsibilities:

- Design and develop the user interface using React.js.
- Create responsive pages and reusable components.
- Develop application pages including:
 - Login page
 - Registration page
 - Dashboard
 - Stock listing page
 - Portfolio page
- Implement navigation between different modules using React Router.

- Integrate frontend components with backend APIs.
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Member 2: _____

Role: Backend Developer

Responsibilities:

- Develop the server-side application using Node.js and Express.js.
 - Create and manage REST APIs required for the application.
 - Implement user authentication and authorization.
 - Develop APIs for:
 - User registration
 - User login
 - Stock data handling
 - Buying stocks
 - Selling stocks
 - Portfolio management
 - Implement backend logic for paper trading operations.
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Member 3: _____

Role: Database Developer

Responsibilities:

- Design and manage MongoDB database structure.
 - Create and maintain database collections.
 - Develop schemas using Mongoose.
 - Manage storage of:
 - User details
 - Stock information
 - Portfolio data
 - Transaction history
 - Ensure proper connection between backend and database.
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Member 4: _____

Role: Stock Data and Trading Module Developer

Responsibilities:

- Integrate stock market API for fetching US stock information.
 - Implement stock search and stock details functionality.
 - Handle historical stock data retrieval.
 - Implement stock chart visualization.
 - Develop paper trading features:
 - Virtual balance management
 - Buy stock functionality
 - Sell stock functionality
 - Portfolio updates after transactions
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Member 5: _____

Role: Integration, Testing and Deployment Developer

Responsibilities:

- Integrate frontend, backend, and database modules.
 - Test complete application workflow.
 - Identify and fix application errors.
 - Perform testing of:
 - Authentication
 - Stock display
 - Trading operations
 - Portfolio management
 - Manage project repository using GitHub.
 - Handle application deployment.
 - Prepare project demonstration and final submission.
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13. Future Enhancements

Possible improvements:

- Real-time stock price updates using WebSockets
 - AI-based stock analysis
 - Advanced trading strategies
 - Mobile application
 - Multiple portfolio support
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14. Conclusion

SB Stocks provides a simple and effective platform for users to understand stock trading through a simulated environment. The application combines stock market data, paper trading functionality, and portfolio management to create an interactive learning experience.